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ORIGINAL ARTICLE

Blood pressure percentiles by age and height for children and adolescents in Tehran, Iran

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To construct reference percentiles for blood pressure (BP) by sex, age and height for the first time in Iran, we used data on 16 972 healthy children, aged 1 month to 18 years, collected during 2000–2010 in Tehran. BP in this population rose steadily with age and height following a very similar trend in both genders up to the age of 14. Systolic BP (SBP) rise was more prominent in younger ages, and after puberty (15–18 years) was greater in boys compared with girls, while the rise in diastolic BP (DBP) was slightly higher in girls. Iranian norms, compared with 'Fourth Report on the Diagnosis, Evaluation, and Treatment of High Blood Pressure in Children and Adolescents' (US-4th-Report) and the 'German BP Percentiles by Age and Height for Children and Adolescents' (KiGGS), showed a similar pattern of differences for both genders. For example, for Tehrani boys up to 6 years old whose heights were equal to 50th percentile of stature-for-age as well as length-for-age growth charts, the differences in 95th percentile for SBPs compared with the US-4th-report varied from 2–21 mm Hg while compared with KiGGS, maximum of differences was 9 mm Hg. For boys 7–15 years of age, ours were slightly higher than both. For ages of 16 and 17 years, we yielded figures lower than US-4th-report (2 mm Hg) but higher than KiGGS (3 mm Hg). Iranian 95th percentile for DBPs was lower than US-4th-report and KiGGS (1–11 mm Hg). Considering the differences with US-4th-report and KiGSS standards, the references presented in this study should rather be applied in Iranian population.

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INTRODUCTION

High blood pressure (BP) is one of the most common risk factors for cardiovascular disease, which is a major cause of disability and premature death throughout the world.[1] Therefore, evaluation of BP and prevention of hypertension in children and adolescents has become a global priority issue in health policies.[2] BP levels in childhood and adolescence are predictive of adult hypertension.[3] Thus, measurement of BP is considered to be an integral part of routine physical examination in childhood, and should be interpreted according to the standard BP nomograms.[4] Local reference data are essential for evaluating observed BP levels. To establish a standard definition for BP in children from birth to 18 years, a wide range of studies based on age, height and weight regarding BP values was reported by Task Force on BP Control in Children in the US.[5,6] However, the distributions of BP values and prevalence of hypertension varies among different racial and ethnic groups.[7,8] Based on these differences, standard values developed for one population might not be applicable to others.[8,9] To date, there have been no valid standard references for BP measurements for Iranian children and adolescents. Although a few studies were conducted to calculate the references for SBP and DBP measurements, none of them represented the whole age groups of children and adolescents and they were not body size adjusted or nationally representative.[10–13]

To our knowledge, there are no BP reference data on Iranian children and adolescents based on a nationally representative survey. This study was designed to provide age–height-related BP reference standards for all age groups of Iranian children and adolescents. The BP values of Iranian children were also compared with the references from the developed world.

MATERIALS AND METHODS

Study population

Analysis was performed on data gathered from 16 972 children and adolescents from Tehran assumed to be representative of the total Iranian pediatric population.[14] Data from three cross-sectional studies were used to construct reference percentiles for BP measurements by sex and age (from 1 month to 18 years). In the first study, which was carried out from November 2000 to November 2002, 8848 primary school children aged 7–12 years from Tehran were included. The second study recorded BP measurements for 6017 children and adolescents aged 12–18 years from secondary and high schools of Tehran from February to September 2004. The latest study in 2010 included BP values for 2107 children younger than 7 years old. In this study, infants (1 month up to 2 years old) were randomly selected from health centers and together with children (older than 2 years) from kindergartens, using a two-stage cluster sampling method.[15] In order to prevent including one subject more than once, we carefully compared the subjects in these three data sets, by identifying their full names and national identity.

Inclusion and exclusion criteria

In all groups BP was measured in a randomized sample from 20 different geographical areas of Tehran. The criteria for selecting a healthy child are: absence of fever or documented disease at the time of BP measurments;

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no evidence of significant hemodynamic problems such as congenital heart disease, or prior cardiac catheterization or thoracic surgery; not taking any medication such as antihypertensive at the time of the study. Data were collected by trained interns of Tehran University of Medical Sciences. The study protocol was approved by Research Ethics Committee of Tehran University of Medical Sciences (TUMS) of Iran. Informed consent was obtained from parents or guardians.

Measurements

BP measurements. All BP measurements were performed with a standard mercury sphygmomanometer (Model 1002/Presameter, Riester, Germany) with the child in a comfortable sitting position, after a period of 5-min rest. The mercury sphygmomanometer remains the 'gold standard' device for BP measurement.[16] For infants, BP was measured in supine position. Measuring BP in infants is difficult mainly due to their unstable posture and tendency to cry, so our trained interviewers took appropriate measures to calm the infant and perform a standard examination. This is also what makes this study unique.

The right arm was positioned at heart level. An appropriate size of cuff was selected using a bladder width ~40% of the arm length and long enough to cover 80–100% of the arm circumference. The cuff was placed on the arm leaving the antecubital fossa free for auscultation, and inflated to a level that occluded the pulse at the wrist. The stethoscope was placed over the brachial artery and the cuff was deflated. The onset of the first Korotkoff (K1) sound was used as a measure for systolic BP (SBP). It has been observed that in younger children the sound can be heard through 0 mm Hg, therefore less pressure was applied to the head of the stethoscope in these cases. K4 diastolic BPs (DBPs) were used as the standards for infants, toddlers and for children aged 3–12 years. K5 diastolic BPs were used as the standards for adolescents aged 13–18 years.

For each child, BP was measured twice with an interval of at least 1 min, and the average of these two recorded measurements was used for data analysis. The differences between the two measurements were negligible. For boys, in age groups of 1–2, 3–6, 7–12 and 13–18 years, the mean of absolute deviation between SBP and DBPs were 1.15, 1.48, 0.18, 0.30 mm Hg and 1.03, 1.12, 0.22, 0.26 mm Hg, respectively. These differences for girls were 1.25, 1.58, 0.49, 0.30 mm Hg and 0.96, 1.24, 0.56, 0.41 mm Hg, respectively. Further details regarding the methods were published in our previous studies.[11,15,17]

In addition, because of variability in BP, to make sure that the recordings were consistent in hypertensive subjects, measurements were repeated twice with 1 week intervals in our second data set for the children aged 12–18 years with BPs ⩾95th percentile. Of the 6017 children in our second data set, 287 (4.7%) subjects were categorized as hypertensive in the first screening which decreased to 62 (1%) in the second screening and finally reached to 46 (0.8%) subjects in the last evaluation. Also, in our data for 2017 children under 7 years old, 102 (4.8%) were categorized as being hypertensive in the first screening which decreased to 5 (0.2%) in the last evaluation.

Height and weight measurements. To assess the infants' weight with accuracy of 50 grams, SECA baby scale model 725, Los Angeles, CA, USA, was used, and their recumbent length with accuracy of 1 mm was measured using SECA mobile measuring mat model 210, USA. Weight of children (2–6 years old) was measured with an accuracy of 500 g, using SECA scale model 760, USA, and to assess their standing height with an accuracy of 1 mm, SECA mechanical measuring tape model 206, USA, was used.

In primary, secondary and high schools, height (cm) without shoes was measured using a stadiometer (SECA Model 207, Hamburg, Germany) with the student standing upright with the heels and back against a vertical scale. Weight without shoes and heavy outer clothing was measured on a balanced scale (SECA Model 710, Germany) that was calibrated daily.

Age-, height- and sex-specific BP level nomograms. Four separate models for BP percentiles (SBP and DBP of boys and girls) as a function of age and height were constructed. First, for both genders height-for-age growth charts were modeled using the Cole's LMS method[18] which summarizes the changing distribution by three smoothed curves; L (for lambda), M (M for mu) and S (S for sigma) representing the skewness (lambda for L parameter), the median (mu for M parameter) and coefficient of variation (sigma for S parameter). Then, the BP reference curves for age and height were fitted simultaneously using an extension of the LMS method for two covariates, the generalized additive models for location scale and shape (GAMLSS) along with the Box-Cox Cole and Green distribution family.[19–21] Free statistical software R 2.15.2 (http://www.R-project.org) was used with GAMLSS 4.2-0. Neither of the models required assumption of normality or a constant variance of BP values with age and/or height. The L, M and S parameters have been modeled as a function of age and/or height either as polynomials or nonparametrically by cubic splines. Goodness of fit was examined via the generalized Akaike information criterion and by examining the percentage of data outside the smoothed percentiles.

Height-for-age LMS parameters for boys and girls were modeled and computed: 3, 5, 5 and 3, 5, 4, respectively.

Median of SBP varied with age and height. Median of DBP was given by a quadratic function in age and linear in height. Model of median SBPs for both genders included cubic spline for age and linear for height and its interaction terms (for example, height × age and height × age^2). Similar models were used for the median of DBP in boys and girls. However for girls' DBP the interaction term of height × age^2 was excluded as it was not significant. Coefficient of variation (S) for both genders varied as a linear function of age. The skewness parameter (L) of SBP and DBP for boys were 1.57 and 2.30 and for girls were 1.17 and 2.13, respectively.

Initially the mentioned percentiles were reported in this study, but any P_α can be calculated by

$$P_\alpha = M(1 + LSZ_\alpha)^{1/L}; L \neq 0$$

Or

$$P_\alpha = M \times \exp(SZ_\alpha); L = 0$$

Where P is the physical measurement and P_α is the 100α percentile, Z_α is the α quantile of a standard normal distribution.

Age in years was used as a continuous variable. Then, the values of the percentiles were calculated by using age in years as 0.5, 1.5, 2.5 and so on, but in Tables 2 and 3 the values are shown in complete years (1, 2, 3, …, 18). The calculations were conducted using R software.

Finally, we compared the fitted percentile curves with the US reference values presented in the 'Fourth Report on the Diagnosis, Evaluation, and Treatment of High Blood Pressure in Children and Adolescents' (US-4th-Report) and the German BP Percentiles by Age and Height for Children and Adolescents (KiGGS).[22]

RESULTS

Using three cross-sectional studies, SBP and DBP levels of 16 972 Iranian healthy children (8787 boys and 8185 girls) from birth to 18 years of age were measured during 2000–2010. Baseline characteristics of the reference population are shown in Table 1. Smoothed BP percentiles of boys and girls by age and height are presented in Tables 2 and 3. As the height-for-age percentiles for are given in centimeters (cm) in order to use BP references, one does not require additional height-for-age reference tables for both genders. As the height percentiles for each age are given in centimeters (cm), in order to use these BP references, one does not require additional height-for-age reference tables for both genders. SBP and DBP in infants, toddlers, children and adolescents aged 1 month to 18 years rose steadily with age and height. SBP rise was more prominent in younger ages and after puberty (15–18 years) and was greater in boys compared with girls. On the contrary, this rise in DBP was slightly higher in girls. The BP differences between boys and girls were at the most 11 mm Hg for SBP 95th percentile and 3 mm Hg for DBP 95th percentile (for both the 95th height percentile). At a particular age and gender, BP percentiles vary by height. For example, Figure 1 presents how the 95th percentile, which is used as the definition of hypertension in children and adolescents, differ between the 5th and the 95th height percentiles. As can be seen for the 5th and 95th height percentiles, SBP for boys is 5–9 mm Hg and for girls it is 2–10 mm Hg, respectively. As for DBP, these figures are 1–9 mm Hg for boys and 2–7 mm Hg for girls, respectively.

We compared Iranian norms with US-4th-Report and KiGGS BP percentiles all modeled by age and height simultaneously. Figure 2 shows these comparisons for boys and girls for median height according to Centers for Disease Control and Prevention (CDC) growth charts (www.cdc.gov/growthcharts). To do this, we computed the 50th and 95th BP percentiles for our children

Table 1. Baseline characteristics of the reference population of children and adolescents (8787 boys and 8185 girls and aged 1–18 years)

Characteristics	*Age (years)*			
	1–2	*3–6*	*7–12*	*13–18*
Children included, n				
Boys	406	746	4505	3130
Girls	320	635	4698	2532
Weight, mean (s.d.), kg				
Boys	8.36 (2.26)	16.96 (4.06)	29.50 (8.38)	58.33 (15.15)
Girls	8.02 (2.31)	15.86 (3.50)	27.73 (7.92)	48.28 (11.51)
Height, mean (s.d.), cm				
Boys	69.5 (8.7)	103.3 (9.5)	133.2 (10.3)	165.6 (11.5)
Girls	69.3 (9.4)	102.1 (8.8)	131.0 (10.3)	155.3 (8.7)
BMI, mean (s.d.), kg m^{-2}				
Boys	17.07 (2.29)	15.73 (2.07)	16.36 (2.89)	21.05 (4.10)
Girls	16.43 (2.13)	15.12 (2.07)	15.89 (2.85)	19.82 (3.70)
Mean (s.d.) of SBP, mm Hg				
Boys	71.75 (9.75)	93.56 (10.24) (10.61)	107.24 (8.98)	115.35 (11.51) (11.72)
Girls	72.97 (10.45)	91.50 (8.79) (10.61)	107.45 (8.90)	108.97 (10.66) (11.72)
Mean (s.d.) of DBP, mm Hg				
Boys	41.73 (5.55)	55.35 (9.61) (10.61)	63.69 (9.49) (10.61)	72.41 (7.87) (11.72)
Girls	41.94 (6.34)	54.41 (8.48) (10.61)	63.34 (9.57) (10.61)	71.21 (9.11) (11.72)
Overweight and obese, n *(weighted % of 8787 boys and 8185 girls)*				
Boys	68 (16.8)	119 (15.9)	436 (9.7)	572 (18.3)
Girls	40 (12.5)	70 (11.0)	325 (7.9)	288 (11.5)

Abbreviations: BMI, body mass index; BP, blood pressure; DBP, diastolic blood pressure; SBP, systolic blood pressure.

whose heights at any given age were similar to the US-4th-Report, although their statistical models differed from ours. The corresponding KiGGS BP percentiles were read off from Figure 3 by Neuhauser *et al.*[22] As we used K4 sound for measuring DBP in subjects younger than 13, for our data to be comparable with the US-4th-Report and KiGGS standards which used the K5 sound, we calculated approximately equivalent values by subtracting 10 mm Hg from our DBP measurements (as K5≅K4-10[23,24]). Generally, the pattern of differences for boys and girls was similar. For boys younger than 4 years with median CDC heights, the US-4th-Report 50th SBP was 1–13 mm Hg higher than ours but the differences were at the most 4 mm Hg compared with KiGGS. Between the ages 5 and 15 years, our BP percentiles were higher than US-4th-Report and KiGGS (6 mm Hg at the most). Although for the subjects aged 16 and 17 years old, Iranian 50th percentile was higher compared with both references, with a maximum difference of 2 mm Hg compared with US-4th-Report which is lower than its maximum difference compared with KiGGS (7 mm Hg). For DBP up to the age of 10 years, our 50th percentile was lower than US-4th-Report and KiGGS (5 mm Hg at the most). These differences were higher in older ages up to 17 years, and ranged from 1 to 9 mm Hg.

The differences in 95th SBP percentile compared with US, varied from 2 to 21 mm Hg up to 6 years but with KiGGS were at the most 9 mm Hg. Between 7 and 15 years, ours were slightly higher than both. And for the ages of 16 and 17 years was lower than US-4th-Report (2 mm Hg) but higher than KiGGS (3 mm Hg at the most). Iranian 95th percentile of DBP was lower than US-4th-Report and KiGGS (ranged from 1 to 11 mm Hg).

For girls with median CDC heights, up to the age of 4 years, the US 50th SBP was 1–15 mm Hg higher than ours but KiGGS was 5 mm Hg higher at the most. Between the ages 5 and 15 years, our BP percentile was higher than US-4th-Report and KiGGS (6 mm Hg at the most). For the ages 16 and 17 years, our 50th percentile was slightly higher, but similar to KiGGS. For DBP up to the age of 10, our 50th percentile was lower than US-4th-Report and KiGGS (6 mm Hg at the most). These differences were higher in older ages up to 17 years and ranged from 1 to 9 mm Hg. The differences in 95th SBP percentile compared with US-4th-Report, varied from 2 to 23 mm Hg up to the age of 6 years but with KiGGS were 19 mm Hg at the most. Between the ages 7 and 13 years, ours were slightly higher than both. On the other hand for the subjects aged 14 to 17 years, ours were lower than US-4th-Report (2 mm Hg) but similar to KiGGS. Our 95th percentile of DBP were lower than US-4th-Report and KiGGS up to the age of 13 years (ranged from 1 to 13 mm Hg) and higher than both (4 mm Hg at the most) from 14 to 17 years.

DISCUSSION

Hypertension is a major public health problem throughout the world, because of its high prevalence and association with high morbidity and mortality.[17] Due to the fact that adult BP levels can be predicted according to childhood BP levels, many epidemiological studies in different geographical areas have been performed to identify normal standard references of BP in childhood in relation to age, sex and body size.[25,26] To define standard references for pediatric BP, the American Academy of Pediatrics Task Force on Blood Pressure published a series of wide range reports on BP levels in relation to age, height and weight from birth to 18 years.[8,9] Recently, Neuhauser *et al.*[22] reported a new standard reference for non-overweight children and adolescents in Germany using new statistical methods, GAMLSS, including concurrent effects of age and height on BP measurements.

This study presents BP references by age and height simultaneously, for infants, toddlers, children and adolescents aged 1 month to 18 years. These findings are based on BP measurements using three nationally representative samples of 16 972 children in Tehran, Iran. Improved statistical methods were used for percentile derivation.

Neuhauser *et al.*,[22] unlike our survey and the US-4th-Report BP references which are widely used, excluded the overweight children when deriving references from KiGGS data. In a reanalysis of the US-4th-Report data excluding overweight children,[27] prehypertension thresholds were only 1–3 mm Hg lower than that of the US-4th-Report.

To investigate the impact of not excluding the overweight children from our reference population, we compared presented BP percentiles with those calculated with excluding overweight children (all other exclusion criteria being the same). Excluding overweight and obese children (⩾85th CDC body mass index percentile) showed negligible differences in results. We used the definition 'equal or greater than the 85th CDC BMI percentile' for infants from 1 month up to 2 years and children 2–18 years old for overweight and obesity according to the Camden's Clinical Commissioning Group.[28] For example, in boys, the average

Table 2. BP levels for boys according to age and height

Age (year)	Height (cm)	SBP (mm Hg)				DBP (mm Hg)			
		50th Percentile (median)	90th Percentile	95th Percentile	99th Percentile	50th Percentile (median)	90th Percentile	95th Percentile	99th Percentile
1	56	67	75	77	80	37	44	46	49
	59	68	76	78	82	38	46	47	51
	63	70	78	80	84	40	47	49	52
	67	72	80	82	86	41	49	51	54
	71	73	81	84	88	42	51	53	56
	74	75	83	85	90	44	52	54	58
	76	75	84	86	91	44	53	55	58
2	68	76	84	86	91	43	51	53	56
	71	76	85	87	92	44	52	54	57
	74	78	87	89	94	45	53	55	59
	78	80	89	91	95	46	55	57	61
	82	81	90	93	97	47	56	58	62
	86	83	92	94	99	49	58	60	64
	88	83	93	95	100	49	58	61	65
3	79	83	92	94	99	48	56	58	62
	81	83	93	95	100	48	57	59	63
	85	85	94	97	102	49	58	60	64
	89	86	96	98	103	51	60	62	66
	93	88	97	100	105	52	61	63	67
	97	89	99	102	107	53	62	65	69
	99	90	100	103	108	54	63	65	70
4	89	88	98	101	106	51	60	62	66
	91	89	99	102	106	52	61	63	67
	94	90	100	103	108	53	62	64	68
	98	91	102	104	109	54	63	66	70
	102	93	103	106	111	55	65	67	71
	106	94	104	107	113	56	66	68	72
	108	95	105	108	113	57	66	69	73
5	97	93	103	106	111	54	63	65	70
	99	93	104	106	112	55	64	66	70
	102	94	105	108	113	56	65	67	71
	106	96	106	109	114	57	66	68	73
	110	97	107	110	116	58	67	70	74
	114	98	109	112	117	58	68	71	75
	116	99	109	112	118	59	69	71	76
6	103	96	107	109	115	56	66	68	72
	105	97	107	110	115	57	66	68	73
	109	98	108	111	117	58	67	69	74
	113	99	110	113	118	59	68	71	75
	117	100	111	114	119	60	69	72	76
	120	101	112	115	121	60	70	73	77
	123	102	113	116	121	61	71	73	78
7	109	99	110	113	118	58	67	70	74
	111	100	110	113	119	59	68	70	74
	115	100	111	114	120	60	69	71	75
	119	102	113	116	121	60	70	72	76
	123	103	114	117	122	61	71	73	78
	127	104	115	118	124	62	72	74	79
	129	104	116	119	124	62	72	75	79
8	115	101	112	115	121	60	69	71	76
	117	102	113	116	121	61	70	72	76
	121	103	114	117	123	61	70	73	77
	125	104	115	118	124	62	71	74	78
	129	105	116	119	125	63	72	75	79
	133	106	117	120	126	64	73	76	80
	135	107	118	121	127	64	74	76	80
9	120	103	115	118	123	62	71	73	77
	122	104	115	118	124	62	71	74	78
	126	105	116	119	125	63	72	74	79
	130	106	117	120	126	64	73	75	79
	135	107	118	122	127	64	74	76	80
	139	108	119	123	128	65	75	77	81
	141	109	120	123	129	65	75	77	82
10	125	105	116	119	125	63	72	75	79
	127	106	117	120	126	64	73	75	79
	131	107	118	121	127	64	73	76	80

Table 2. (Continued)

Age (year)	Height (cm)	SBP (mm Hg)				DBP (mm Hg)			
		50th Percentile (median)	90th Percentile	95th Percentile	99th Percentile	50th Percentile (median)	90th Percentile	95th Percentile	99th Percentile
	135	108	119	122	128	65	74	77	81
	140	109	120	123	129	66	75	77	82
	144	110	121	124	130	66	76	78	82
	146	110	122	125	131	67	76	79	83
11	129	107	118	121	126	65	74	76	80
	132	107	118	121	127	65	74	76	81
	136	108	119	122	128	66	75	77	81
	140	109	120	124	129	67	76	78	82
	145	110	122	125	131	67	76	79	83
	149	111	123	126	132	68	77	79	84
	152	112	123	127	132	68	77	80	84
12	134	108	119	122	128	66	75	78	82
	136	108	119	123	128	67	76	78	82
	141	109	121	124	129	67	76	79	83
	146	110	122	125	131	68	77	79	83
	151	112	123	126	132	69	78	80	84
	156	113	124	127	133	69	78	81	85
	159	113	125	128	134	70	79	81	85
13	139	109	120	123	129	68	77	79	83
	142	110	121	124	130	68	77	79	83
	147	111	122	125	131	69	78	80	84
	152	112	123	127	132	70	78	81	85
	158	113	125	128	134	70	79	81	85
	163	114	126	129	135	71	80	82	86
	166	115	127	130	136	71	80	82	87
14	144	110	121	125	130	70	78	80	84
	147	111	122	125	131	70	79	81	85
	153	112	124	127	133	70	79	81	85
	159	114	125	128	134	71	80	82	86
	164	115	127	130	136	72	80	83	87
	170	116	128	131	137	72	81	83	87
	173	117	129	132	138	72	81	84	88
15	150	112	123	126	132	71	80	82	86
	153	113	124	127	133	71	80	82	86
	159	114	125	129	134	72	80	83	87
	164	115	127	130	136	72	81	83	87
	170	117	128	132	138	73	81	84	88
	175	118	130	133	139	73	82	84	88
	178	119	131	134	140	74	82	85	89
16	156	114	125	128	134	73	81	83	87
	159	114	126	129	135	73	81	83	87
	164	116	127	130	136	73	82	84	88
	169	117	129	132	138	74	82	84	88
	175	118	130	133	139	74	82	85	89
	179	120	132	135	141	74	83	85	89
	182	120	132	136	142	75	83	85	89
17	161	115	127	130	136	74	82	84	88
	164	116	128	131	137	74	82	85	88
	168	117	129	132	138	74	83	85	89
	173	119	131	134	140	75	83	85	89
	178	120	132	135	141	75	83	86	90
	182	121	133	137	143	75	84	86	90
	185	122	134	137	143	76	84	86	90
18	165	117	129	132	138	75	84	86	89
	168	118	130	133	139	76	84	86	90
	172	119	131	134	140	76	84	86	90
	177	120	132	136	142	76	84	86	90
	181	122	134	137	143	76	84	87	90
	185	123	135	138	144	76	85	87	91
	187	123	136	139	145	77	85	87	91

Abbreviations: BP, blood pressure; DBP, diastolic blood pressure; SBP, systolic blood pressure. Height in centimeters for each age represents the 5th, 10th, 25th, 50th, 75th, 90th and 95th percentile.

Table 3. BP levels for girls according to age and height

Age (year)	Height (cm)	SBP (mm Hg)				DBP (mm Hg)			
		50th Percentile (median)	90th Percentile	95th Percentile	99th Percentile	50th Percentile (median)	90th Percentile	95th Percentile	99th Percentile
1	55	67	74	76	80	39	46	48	51
	58	68	75	77	81	40	47	49	52
	62	70	77	80	83	41	48	50	53
	67	72	80	82	86	42	50	52	55
	71	74	81	84	88	43	51	53	56
	74	75	83	85	90	44	52	54	58
	76	76	84	86	91	45	53	55	58
2	68	75	83	85	90	44	51	53	57
	71	76	84	86	91	44	52	54	58
	74	77	86	88	93	45	53	55	59
	78	79	88	90	94	46	54	56	60
	82	81	89	92	96	47	56	58	61
	85	82	91	93	98	48	57	59	63
	87	83	92	94	99	49	57	59	63
3	80	82	91	93	98	48	56	58	62
	82	83	92	94	99	48	57	59	62
	85	84	93	96	100	49	57	60	63
	88	85	95	97	102	50	59	61	65
	92	87	96	99	104	51	60	62	66
	95	88	97	100	105	52	61	63	67
	97	89	98	101	106	52	61	63	67
4	90	88	97	100	105	51	60	62	66
	91	88	98	100	105	51	60	62	66
	94	89	99	102	107	52	61	63	67
	98	90	100	103	108	53	62	64	68
	101	92	102	104	109	54	63	65	69
	104	93	103	106	111	55	64	66	70
	106	94	104	106	112	55	64	67	71
5	97	92	102	105	110	54	62	65	69
	99	93	102	105	110	54	63	65	69
	102	93	104	106	112	55	64	66	70
	105	95	105	108	113	55	65	67	71
	109	96	106	109	114	56	66	68	72
	112	97	107	110	116	57	66	69	73
	114	98	108	111	116	57	67	69	74
6	104	96	106	109	114	56	65	67	71
	105	96	106	109	115	56	65	68	72
	108	97	107	110	116	57	66	68	73
	112	98	109	112	117	57	67	69	74
	115	99	110	113	118	58	68	70	75
	119	100	111	114	120	59	69	71	76
	121	101	112	115	121	60	69	72	76
7	109	98	109	112	118	57	67	69	73
	111	99	110	113	118	58	67	69	74
	114	100	111	114	119	59	68	70	75
	118	101	112	115	121	59	69	71	76
	121	102	113	116	122	60	70	72	77
	125	103	114	117	123	61	71	73	78
	127	104	115	118	124	61	71	74	78
8	114	101	112	115	121	59	68	71	75
	116	101	112	115	121	60	69	71	76
	119	102	113	116	122	60	70	72	76
	123	103	115	118	123	61	71	73	77
	127	105	116	119	125	62	72	74	79
	131	106	117	120	126	63	73	75	80
	133	106	118	121	127	63	73	76	80
9	119	103	114	117	123	61	70	73	77
	121	104	115	118	124	61	71	73	77
	124	105	116	119	125	62	72	74	78
	128	106	117	120	126	63	72	75	79
	133	107	118	121	127	64	73	76	81
	137	108	119	123	129	65	74	77	82
	139	108	120	123	129	65	75	78	82
10	124	105	116	119	125	63	72	74	79
	126	106	117	120	126	63	73	75	79
	130	106	118	121	127	64	73	76	80

Table 3. (Continued)

Age (year)	Height (cm)	SBP (mm Hg)				DBP (mm Hg)			
		50th Percentile (median)	90th Percentile	95th Percentile	99th Percentile	50th Percentile (median)	90th Percentile	95th Percentile	99th Percentile
	135	108	119	122	128	65	74	77	81
	139	109	120	124	130	66	76	78	83
	143	110	121	125	131	66	76	79	84
	146	110	122	125	132	67	77	80	84
11	129	107	118	121	127	64	74	76	81
	131	107	119	122	128	65	75	77	81
	136	108	120	123	129	66	75	78	82
	141	109	121	124	130	67	76	79	84
	146	110	122	125	131	68	78	80	85
	150	111	123	126	133	68	78	81	86
	153	112	124	127	133	69	79	82	86
12	134	108	119	123	129	66	76	78	83
	137	108	120	123	129	67	77	79	83
	142	109	121	124	130	68	77	80	84
	147	110	122	125	132	69	78	81	86
	152	111	123	127	133	69	79	82	87
	156	112	124	128	134	70	80	83	88
	159	113	125	128	134	71	81	84	88
13	140	109	120	124	130	68	78	80	85
	143	109	121	124	130	69	78	81	86
	147	110	122	125	131	70	79	82	86
	152	111	123	126	132	70	80	83	88
	157	112	124	127	134	71	81	84	89
	161	113	125	128	134	72	82	85	89
	164	113	125	129	135	72	83	85	90
14	145	110	121	125	131	70	80	82	87
	147	110	122	125	131	70	80	83	87
	152	111	123	126	132	71	81	84	88
	156	111	123	127	133	72	82	84	89
	161	112	124	128	134	73	83	85	90
	165	113	125	128	135	73	83	86	91
	167	113	125	129	135	74	84	86	91
15	148	110	122	125	131	71	81	84	88
	150	110	122	126	132	72	82	84	89
	154	111	123	126	132	72	82	85	89
	159	112	124	127	133	73	83	86	90
	163	112	124	128	134	74	84	86	91
	167	113	125	128	135	74	84	87	92
	169	113	125	129	135	75	85	87	92
16	150	110	122	125	132	73	82	85	89
	152	111	122	126	132	73	83	85	90
	156	111	123	126	133	73	83	86	90
	160	112	124	127	133	74	84	86	91
	164	112	124	128	134	75	85	87	92
	168	113	125	128	135	75	85	88	92
	170	113	125	129	135	75	85	88	93
17	151	111	122	126	132	74	83	86	90
	153	111	123	126	132	74	84	86	91
	157	111	123	126	133	74	84	87	91
	161	112	124	127	133	75	85	87	92
	165	112	124	128	134	75	85	88	92
	168	113	125	128	134	76	86	88	93
	170	113	125	128	135	76	86	89	93
18	151	111	123	126	132	75	84	87	91
	153	111	123	126	132	75	84	87	91
	157	111	123	127	133	75	85	87	92
	161	112	124	127	133	76	85	88	92
	164	112	124	128	134	76	86	88	93
	168	113	125	128	134	76	86	89	93
	170	113	125	128	135	77	87	89	94

Abbreviations: BP, blood pressure; DBP, diastolic blood pressure; SBP, systolic blood pressure. Height in centimeters for each age represents the 5th, 10th, 25th, 50th, 75th, 90th and 95th percentile.

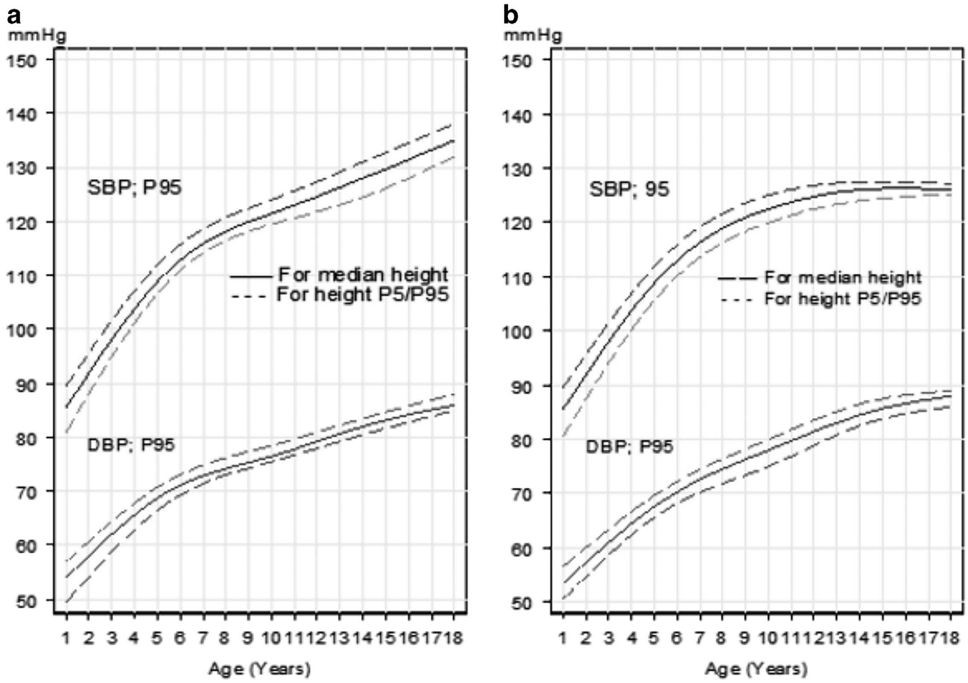

Figure 1. The 95th percentile of SBP and DBP of Iranian boys and girls aged 1–18 years according to age and 50th (solid middle line), 5th (dashed lower line) and 95th (dashed upper line) percentiles of height. A, Boys; B, Girls. P_n indicates n[th] percentile. A full color version of this figure is available at the *Journal of Human Hypertension* journal online.

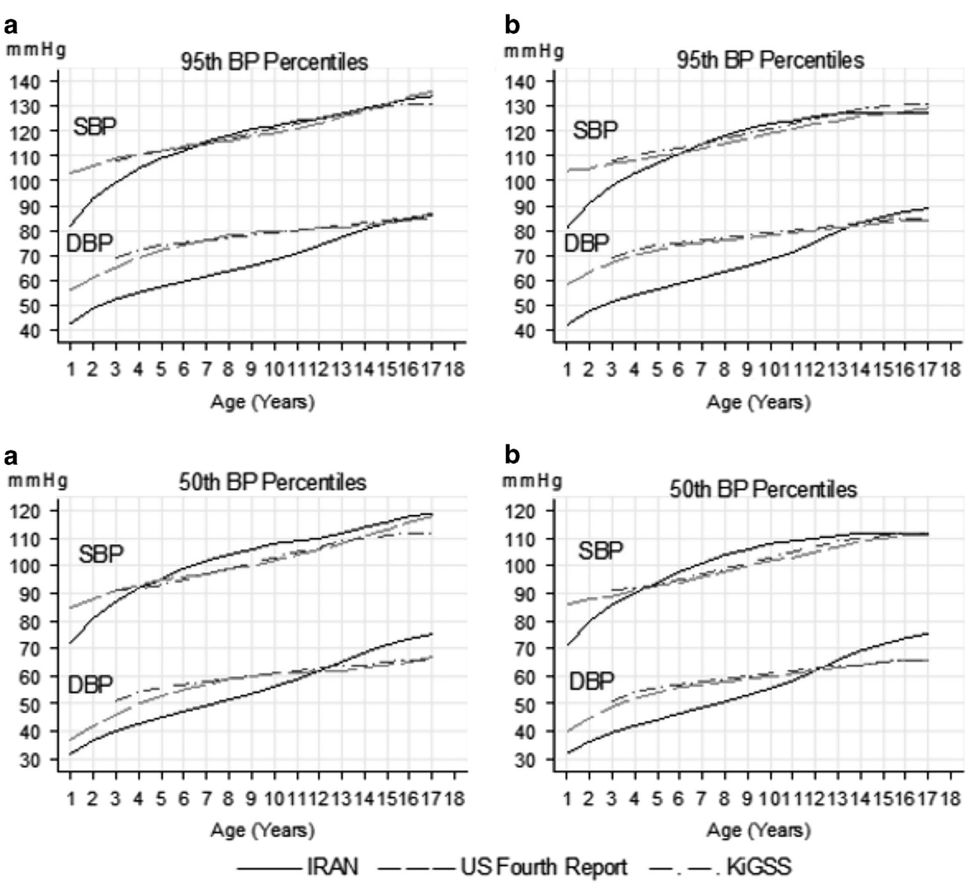

Figure 2. Comparison of SBP and DBP references of Iranian boys and girls with Fourth Report (USA) and KiGGS (Germany) blood pressure references for median height, values, according to Centers for Disease Control and Prevention (CDC). A, Boys; B, Girls. A full color version of this figure is available at the *Journal of Human Hypertension* journal online.

difference between the 50th, 90th, 95th and the 99th percentiles for SBP and DBP for 50th CDC height-for-age percentile, ranged from 0.54 to 1.43 mm Hg. In girls, these differences ranged from 0.44 to 0.92 mm Hg. These differences for the 95th percentiles of height ranged from 0.55 to 1.4 mm Hg and from 0.44 to 0.92 mm Hg for boys and girls, respectively (data are not presented but it was similar). Therefore alike US-4th-Report we presented our BP percentiles for all the subjects including the overweight children.

Moreover for computation of corresponding body mass index values we used WHO standards. To diagnose an individual with hypertension, a few millimeters of Hg may not be relevant, as measurements on multiple occasions are needed to make a diagnosis and also both inter-individual variation and inaccuracy of the BP measuring devices may be greater than that small difference detected.

BP references including age, gender and height simultaneously are scarce. In addition to the US-4th-Report research and the non-overweight KiGGS,[22] Norwegian BP percentiles by age and height have been reported among non-overweight adolescents aged 13–18 years.[29] However, their findings were not only considerably higher compared with ours (for example, for same height values of boys and girls, the 50th and 95th SBP differences were 10.4 and 17.6 mm Hg, respectively), but also compared to the KiGGS's percentiles from non-overweight children and US-4th-Report references in which overweight subjects were not excluded.

British BP references were only modeled with age and for subjects 4–24 years old.[30] To be able to compare our data with them, we modeled our BP measurements only with age. When we compared UK's 50th and 90th BP percentiles for ages 4–18 years with ours, we found that on average UK's SBP percentiles for boys and girls were 5.4, 6.9 mm Hg and 5.8, 7.5 mm Hg (s.d. ~4 mm Hg) higher than ours. However, after adjusting K5 and K4 differences, on an average their 50th and 90th DBP percentiles were at the most about 1.5 mm Hg higher than Iranians (s.d. ~8 mm Hg). Sung *et al.*[31] published BP references by age and height for Hong Kong Chinese children aged 6–18 years. They included overweight children and their results were quite similar to the KiGGS overall sample, but compared with the KiGGS references derived from the non-overweight population, their figures were mostly higher. In general their results were also higher than ours as well. Hong Kong's 95th SBP percentiles for 6- to 18-year old boys and girls were 2–4 mm Hg and 1–7 mm Hg higher than ours, respectively. The 50th SBP percentile for 6- to 12-year old Iranian boys and girls were 2 and 4 mm Hg higher, respectively. But from 13 to 18 years was on an average 3–1 mm Hg lower.

Also, Hong Kong's 95th and 50th DBP percentiles for 6- to 18-year old boys and girls were higher than ours, with the differences declining as the age increases. The boys' 95th DBP percentile was 12–10 mm Hg higher than ours. This figure was 13–5 mm Hg for girls. The 50th DBP percentile for boys was 10–6 mm Hg higher than ours and for girls it was 12–2 mm Hg higher than ours.

Only in one study which was conducted in Shiraz, southern Iran, BP was modeled with age and height for healthy school children aged 6–11 years.[4] When we compared our references with these data, the range of our same height percentiles was wider than Shiraz's children for aged 6–11 years old although the median heights were similar across ages. Also, our SBP percentiles were higher and these differences were more pronounced for 90th and 95th percentiles. They used K5 when measuring DBP so when we adjusted this, the corresponding difference was lower compared with SBP. One reason for the observed differences could be the fact that data obtained from Tehran included a mixed population of different ethnic groups from all over Iran.

The major strength of the Tehran's BP reference is the large and nationally representative sample covering a wide range of subjects from 1 month to 18 years. Application of standardized measurements for BP and height and using a BP measuring device validated in children were other advantages of this study. Calculating the average of two BP measurements for each subject also decreased the measurement bias in our survey. Moreover, modeling by age and height simultaneously with flexible statistical techniques that do not impose normality or constant variance assumptions on the data enriched our findings.

One limitation of our study is that as the BP levels were measured by human, some values (such as those ending by zero or five) had relatively higher frequencies than others, a phenomenon which is called end-digit preference.[32] This may not only drop the goodness of fit criteria, but also can change the fitted models as well as the prevalence of hypertension based on these fitted curves.[33] However, to obtain better fits and reduce the effect of such values, the adjusted model fit was assessed by generalized Akaike information criterion and by examining the percentage of data outside the smoothed percentiles.

Also, our study only measured the BP levels of children and adolescents in Tehran and it might be argued that our findings are not nationally representative, but in developing countries, capital cities (like Tehran) tend to include people drawn from all over the country that can often be assumed to be representative of the entire country's population, as advocated by Hosseini *et al.*[14] in their survey, conducted on the growth of children in Iran.

At the time of proposing this study in 2000 to the vice-chancellor for research of Tehran University of Medical Sciences, pursuing the American Second Task Force on Blood Pressure Control in Children[5] was common practice in Iranian health system. Second Task Force recommended utilizing the fourth Korotkoff phase (K4) for DBP in infants and children 3–12 years old. K5 DBP was recommended to be used for adolescents 13–18 years of age. Others have also used these recommendations in their studies.[3] However, according to the Update on the 1987 Task Force Report on High Blood Pressure in Children and Adolescents, the fifth Korotkoff phase (K5) can also be used to define DBP in both children and adolescents (3–12 years).

It should be noted that since the supervision team was the same in all the three surveys and as stated in methods section (BP measurements), similar measurement method and measuring devices were used, there is no selection and measurement bias. Moreover, although the difference between the times the studies were conducted may induce 'statistically' significant differences in anthropometric characteristics such as height and BP, but these changes were evaluated and found to be negligible. For example, according to Azizi and colleagues[34] the maximum change in height, SBP and DBP of children aged 10–19 years during 1999–2008 in Tehran were 3 cm, 6 mm Hg and 5 mm Hg, respectively. In addition, our previous study[35] showed that the average changes of height in urban boys and girls aged 2–18 years, were 3.4 and 3.6 cm, respectively.

As stated for BP measurement, the mercury sphygmomanometer remains the 'gold standard' device. However, due to environmental concerns in many practice settings (for example, having been banned in Veterans Administration Hospitals), they are being replaced by other types of devices (for example, aneroid sphygmomanometers and digital electronic pressure transducers). On the other hand, there is controversy regarding the accuracy of BP measurement using automated devices. Evidence has shown that automated devices tend to overestimate both SBP and DBP in children and adolescents 5–17 years of age and underestimate both SBP and DBP in adults.[16]

CONCLUSION

The references presented here are, to our knowledge, the first Iranian BP references by age and height covering subjects aged

1 month to 18 years. With respect to US and KiGGS standards, our study implied that BP curves for Iranian children and adolescents have significant differences with other BP levels of a given population; thus the reference standards represented in this study are more applicable among Iranian children and adolescents.

What is known about topic

- Due to the insufficient international data and percentile derivatives by age and height simultaneously, current US references of pediatric blood pressure (BP) are widely used all around the world. However, these references may not be applicable to other populations. Moreover with improved statistical methods we can analyze the new data with greater precision.

What this study adds

- This study gives the BP references by age and height for children and adolescents aged 1–18 years in Tehran, Iran. The study population is a national representative sample, and analysis was done by improved statistical methods. These factors make this study unique and different from the previous surveys in Iran.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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